

Math 252Z - Calculus II - Midterm Exam 1

CRN: 13210

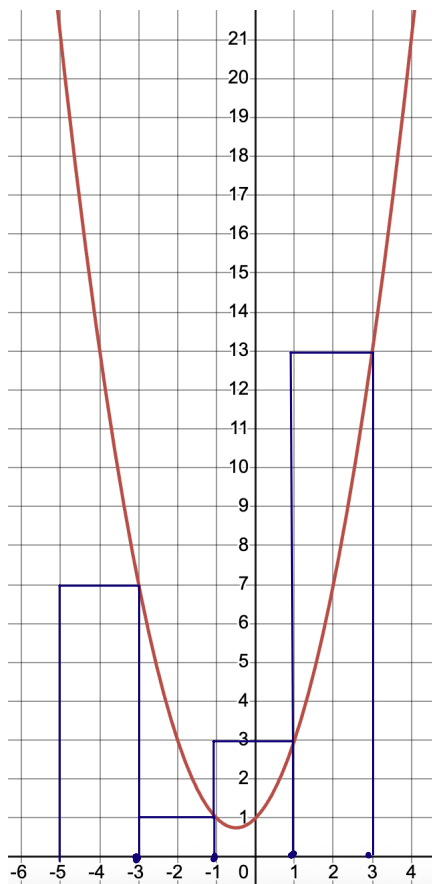
October 24, 2025

Last Name: _____, First Name: _____

1. There are 6 problems. Many problems have multiple parts. The total credit is 100.
2. No collaboration is allowed for this exam. The work on this exam is to be yours and yours alone. Failure to adhere to this policy will result in a zero on this exam. Any instances of suspected cheating will also be reported to the Dean of Students' Office.
3. No calculators, books or other material are permitted.
4. In order to receive full credit you need to show your work, and present your answers as clearly as possible.
5. Answers must be exact (like $\sqrt{2}$ or π), not approximate (like 1.414, 3.14), unless the problem specifically indicates otherwise.
6. Simplify where appropriate. Quantities like $\sqrt{9}$, $\ln e$, inverse trigonometric functions at special values, or trigonometric functions at special angles should be calculated. Reduce all fractions.
7. If an answer line is provided for the answer, place your final answer in it. Otherwise, circle, box, or underline each final answer as appropriate for clarity.
8. Failure to follow directions specific to a problem will result in loss of points.
9. Put away and **silence** your cell phones, tablets, laptops, etc.
10. Once this exam begins, you will have 50 minutes to complete your solutions.

1. [10 pts] Consider the function $f(x) = x^2 + x + 1$ on the interval $[-5, 3]$. The graph of the function is given below.

- 1). Illustrate the right-endpoint Riemann sum R_4 on the interval $[-5, 3]$ with $n = 4$ by sketching the appropriate rectangles on the graph.
- 2). Calculate the right-endpoint Riemann sum R_4 on the interval $[-5, 3]$ with $n = 4$.



$$\Delta x = 2$$

Right endpoints: $-3, -1, 1, 3$

Heights: $f(-3) = 7,$

$$f(-1) = 1$$

$$f(1) = 3$$

$$f(3) = 13$$

$$R_4 = 7 \cdot 2 + 1 \cdot 2 + 3 \cdot 2 + 13 \cdot 2 = 48$$

2. [7 pts] Consider the function $f(x) = x^2 \cot x$ on the interval $[2, 4]$.

- a). If the interval $[2, 4]$ is divided into n subintervals of equal length, what is the width of each subinterval?

Answer: $\Delta x = \frac{2}{n}$

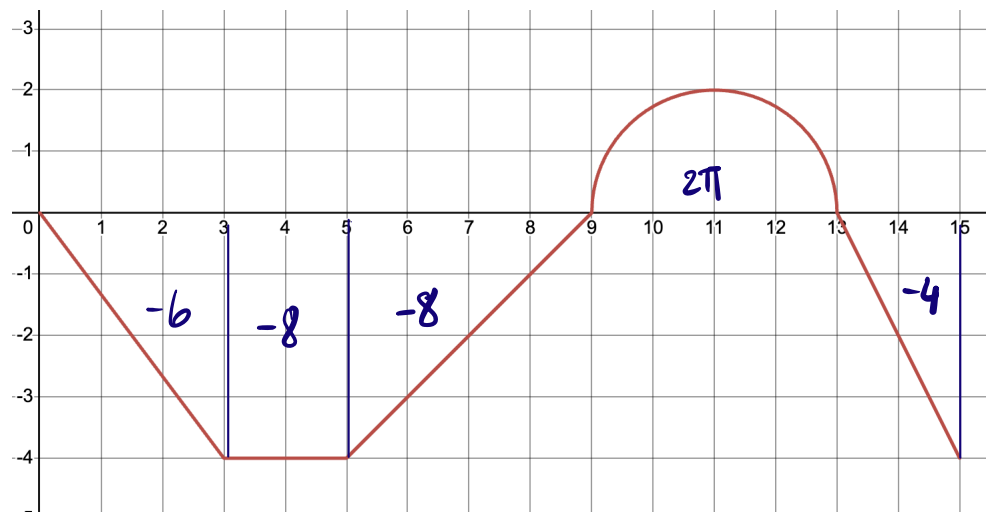
- b). Find the right endpoint x_i of the i th subinterval, and then find $f(x_i)$.

Answer: $x_i = 2 + \frac{2i}{n}, f(x_i) = \left(2 + \frac{2i}{n}\right)^2 \cot\left(2 + \frac{2i}{n}\right)$

- c). Use your work for Part a) and b) to express $\int_2^4 x^2 \cot x \, dx$ as a limit of Riemann sums with right endpoints. Note: You do not need to simplify or evaluate the limit.

Answer: $\int_2^4 x^2 \cot x \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 + \frac{2i}{n}\right)^2 \cot\left(2 + \frac{2i}{n}\right) \cdot \frac{2}{n}$

3. [6 pts] Consider the function $f(x)$, whose graph is given below. Evaluate the following integrals.



1). $\int_0^5 f(x) dx = \underline{-14}$;

2). $\int_5^{13} f(x) dx = \underline{2\pi - 8}$;

3). $\int_9^{15} f(x) dx = \underline{2\pi - 4}$

4. [7 pts] Given $F(x) = \int_2^{x^4} \frac{\sin(\sqrt{t})}{1+t} dt$, use Fundamental Theorem of Calculus-Part 1 to find the derivative $F'(x)$.

$$F'(x) = \frac{\sin(\sqrt{x^4})}{1+x^4} \cdot 4x^3 = \frac{\sin(x^2)}{1+x^4} \cdot 4x^3$$

5. [11 pts] 1). Simplify the sum $R_n = \sum_{i=1}^n \left(2 + \frac{9i^2}{n^2}\right) \cdot \frac{3}{n}$. Write your answer as a function of n without any summation signs.

$$\begin{aligned}
 R_n &= \sum_{i=1}^n \left(2 + \frac{9i^2}{n^2}\right) \frac{3}{n} = \sum_{i=1}^n \left[\frac{6}{n} + \frac{27i^2}{n^3}\right] \\
 &= \sum_{i=1}^n \frac{6}{n} + \sum_{i=1}^n \frac{27}{n^3} i^2 \\
 &= \sum_{i=1}^n \frac{6}{n} + \frac{27}{n^3} \sum_{i=1}^n i^2 \\
 &= \frac{6}{n} \cdot n + \frac{27}{n^3} \cdot \frac{(2n^3 + 3n^2 + n)}{6} \\
 &= 6 + \frac{9}{2} \left(2 + \frac{3}{n} + \frac{1}{n^2}\right) \\
 &= 15 + \frac{27}{2n} + \frac{9}{2n^2}
 \end{aligned}$$

- 2). [2 pts] Evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 + \frac{9i^2}{n^2}\right) \cdot \frac{3}{n}$ by using your result from Part 1).

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 + \frac{9i^2}{n^2}\right) \frac{3}{n} = 15$$

- 3). [3 pts] Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 + \frac{9i^2}{n^2}\right) \cdot \frac{3}{n}$ as a definite integral.

$$\Delta x = \frac{3}{n}, \quad a=0, \quad b=3 \quad x_i = \frac{3i}{n} \Rightarrow f(x_i) = 2 + \left(\frac{3i}{n}\right)^2 \Rightarrow f(x) = 2 + x^2$$

$$\text{Answer: } \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 + \frac{9i^2}{n^2}\right) \cdot \frac{3}{n} = \int_0^3 \frac{2 + x^2}{1} dx.$$

6. Evaluate the following integrals. Make sure to simplify all the trigonometric or inverse trigonometric values in the definite integrals, if any.

1). [10 pts] $\int \frac{x^6 - 6}{x^3} dx$

$$= \int \left(x^3 - \frac{6}{x^3} \right) dx = \int (x^3 - 6x^{-3}) dx = \frac{1}{4}x^4 - 6 \cdot \left(-\frac{1}{2}\right) \cdot x^{-2} + C$$

$$= \frac{1}{4}x^4 + \frac{3}{x^2} + C$$

2). [12 pts] $\int_0^{\frac{1}{2}} \left(\frac{6}{\sqrt{1-x^2}} + \frac{1}{2x+1} \right) dx.$

$$= \left[6 \arcsin(x) + \frac{1}{2} \ln|2x+1| \right] \Big|_0^{\frac{1}{2}}$$

$$= \left(6 \arcsin\left(\frac{1}{2}\right) + \frac{1}{2} \ln|2| \right) - \left(6 \arcsin(0) + \frac{1}{2} \ln 1 \right)$$

$$= 6 \cdot \frac{\pi}{6} + \frac{1}{2} \ln 2 - 0$$

$$= \pi + \frac{1}{2} \ln 2$$

$$u = 2x^4 - 3 \quad du = 8x^3 dx \Rightarrow \frac{1}{8} du = x^3 dx$$

3). [10 pts] $\int x^3 e^{2x^4-3} dx$. (Hint: Use u -substitution.)

$$= \frac{1}{8} \int e^u du = \frac{1}{8} e^u + C = \frac{1}{8} e^{2x^4-3} + C$$

4). [10 pts] $\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$. $u = \sqrt{x}$, $du = \frac{1}{2\sqrt{x}} dx \Rightarrow 2 du = \frac{1}{\sqrt{x}} dx$

$$= 2 \int \sin u du$$

$$= -2 \cos u + C = -2 \cos(\sqrt{x}) + C$$

5). [12 pts] $\int x^4 \ln x dx$.

$$u = \ln x \quad dv = x^4 dx$$

$$= \frac{1}{5} x^5 \ln x - \frac{1}{5} \int \frac{1}{x} \cdot x^5 dx \quad du = \frac{1}{x} dx \quad v = \frac{1}{5} x^5$$

$$= \frac{1}{5} x^5 \ln x - \frac{1}{5} \int x^4 dx$$

$$= \frac{1}{5} x^5 \ln x - \frac{1}{5} \cdot \frac{1}{5} x^5 + C$$

$$= \frac{1}{5} x^5 \ln x - \frac{1}{25} x^5 + C$$

Formula Sheet

$$\sum_{i=1}^n c = cn$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$